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Business Context:

Car crashes are a leading cause of injury and death worldwide, and improving vehicle safety is a critical concern for car manufacturers. With advancements in technology and engineering, manufacturers are continuously seeking ways to design safer vehicles to reduce fatalities and severe injuries in the event of a crash. Despite these efforts, understanding the precise factors that contribute to survival in car crashes remains a complex challenge.

The problem arises from the nature of car accidents, where various elements such as impact speed, the use of safety features, the type of collision, and the demographics of the occupants all play significant roles. Each crash is unique, and even minor variations can significantly affect the outcome for the occupants. This complexity necessitates a detailed analysis to identify which factors are most influential in determining survival outcomes.

Solving this problem is essential for:

1. Safety Regulations
2. Design Improvements
3. Public Health
4. Consumer Confidence

Problem Definition:

Over the last year, the Department of Road Transport has witnessed a 15% YoY rise in the number of car crashes happening in urban areas. While they have the causes of the accidents post-facto, it is essential to pre-empt the risk to increase road safety.

With a sample of the historical car crashes over 5 years, and different attributes of the car, along with the occupant relevant to the car crash, the objective is to analyse the data, identify patterns in car crashes, build a predictive model to determine the likelihood of survival in car crashes based on the factors and identify the most critical factors that influence survival outcomes, thereby helping the department come up with necessary safety regulations that must be adopted by all vehicle manufacturers and users.

Data Background:

The data contains the different attributes of car crashes, with the outcome variable being whether the occupant was deceased during the crash or not.

Column Name	Description
caseid	Unique identifier created by combining the population sampling unit, case number, and vehicle number
speed_range	Estimated impact speed categorized as: 1–9 km/h, 10–24 km/h, 25–39 km/h, 40–54 km/h, 55+ km/h
weight	Observation weights representing inverse probability of sampling (used to ensure representative analysis)
seatbelt	Indicates seatbelt usage by the occupant: *none* or *belted*
frontal_impact	Type of collision: *0 = non-frontal, 1 = frontal*
sex	Gender of the occupant: *f (Female)* or *m (Male)*
age_of_occ	Age of the occupant (in years)
year_of_acc	Year in which the accident occurred
model_year	Manufacturing year of the vehicle
airbag	Airbag deployment status: *deploy*, *nodeploy*, or *unavail*
occ_role	Role of the occupant: *driver* or *pass (passenger)*
deceased	Target variable indicating survival outcome: *no (survived)* or *yes (deceased)*

Data Overview:

```
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   caseid          11217 non-null  object
 1   speed_range     11217 non-null  object
 2   weight          11217 non-null  float64
 3   seatbelt        11217 non-null  object
 4   frontal_impact  11217 non-null  int64
 5   sex             11217 non-null  object
 6   age_of_occ      11217 non-null  int64
 7   year_of_acc     11217 non-null  int64
 8   model_year      11217 non-null  int64
 9   airbag          11217 non-null  object
 10  occ_role        11217 non-null  object
 11  deceased        11217 non-null  object
```

Figure 1: Datatypes Overview

	count	mean	std	min	25%	50%	75%	max
weight	11217.00000	431.40531	1406.20294	0.00000	28.29200	82.19500	324.05600	31694.04000
frontal_impact	11217.00000	0.64402	0.47883	0.00000	0.00000	1.00000	1.00000	1.00000
age_of_occ	11217.00000	37.42765	18.19243	16.00000	22.00000	33.00000	48.00000	97.00000
year_of_acc	11217.00000	2001.10324	1.05681	1997.00000	2001.00000	2001.00000	2002.00000	2002.00000
model_year	11217.00000	1994.17794	5.65870	1953.00000	1991.00000	1995.00000	1999.00000	2003.00000

Figure 2: Statistical Summary

- The dataset contains **11,217 rows and 12 columns** with no missing values and no duplicate records, indicating a clean dataset ready for analysis.
- The target variable deceased is **heavily imbalanced** — approximately 89.5% of occupants survived (no) versus 10.5% deceased (yes), which must be accounted for during model evaluation.
- The weight column (observation/sampling weight) has an unusually large range (0 to ~31,694) with extreme outliers, as it represents inverse probability weights rather than a physical attribute.
- There are no missing values and no duplicate records found in the dataset.

Feature Engineering:

Creating veh_usage_duration:

- veh_usage_duration represents the number of years the vehicle had been in use at the time of the accident. It is computed as: (year_of_acc – model_year)

```
veh_usage_duration created.
count    11217.00000
mean         6.92529
std          5.49315
min         -1.00000
25%          2.00000
50%          6.00000
75%         10.00000
max         49.00000
Name: veh_usage_duration, dtype: float64
```

Figure 3: Distribution of veh_usage_duration

- veh_usage_duration ranges from −1 to ~50 years; the small number of negative values (model_year slightly > year_of_acc) likely reflects minor data entry inconsistencies for brand-new vehicles and will be treated as 0-year-old vehicles during preprocessing.
- The median vehicle age at time of crash is approximately 6–7 years, suggesting that the sample captures a broad mix of relatively new and aging vehicles

Exploratory Data Analysis:

Univariate Analysis:

Each variable is visualized using the most appropriate chart type:

- **Categorical / Binary**: Annotated bar charts is used for showing counts and percentages
- **Continuous**: Combined boxplot + histogram has been used with KDE to reveal shape, spread, and outliers

1. Target Variable — deceased:

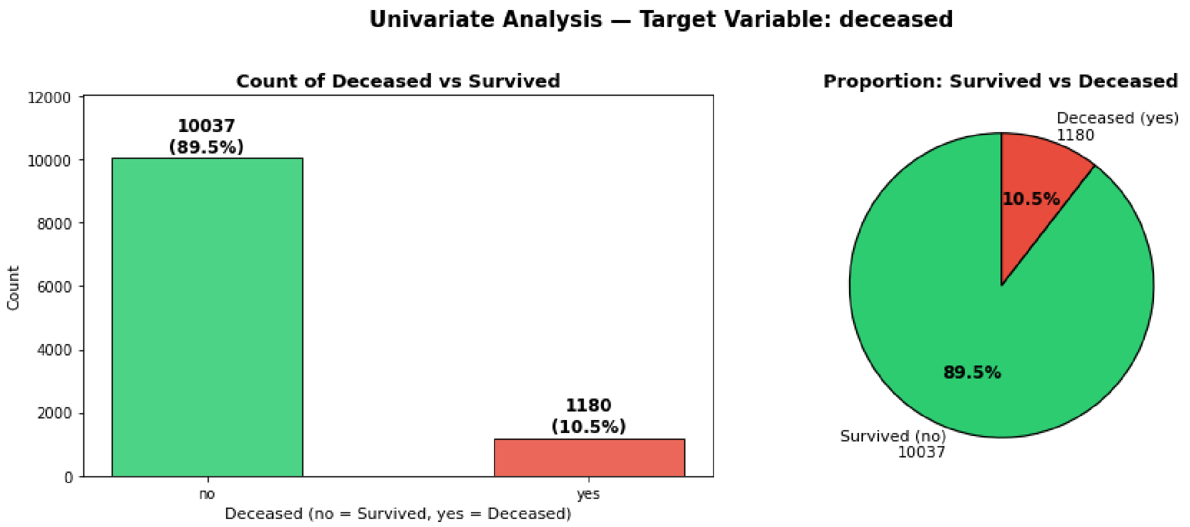


Figure 4: Distribution of deceased (Target Variable)

- The dataset is significantly imbalanced: ~89.5% of occupants survived (no) and only ~10.5% were deceased (yes), with 10,037 surviving versus 1,180 deceased cases.
- This class imbalance implies that a classifier predicting 'survived' for all records would achieve ~89.5% accuracy, making Recall (Sensitivity) the primary metric for evaluation — minimizing false negatives (missed fatality predictions) is critical for road-safety policy decisions.

2. weight — Observation/Sampling Weight

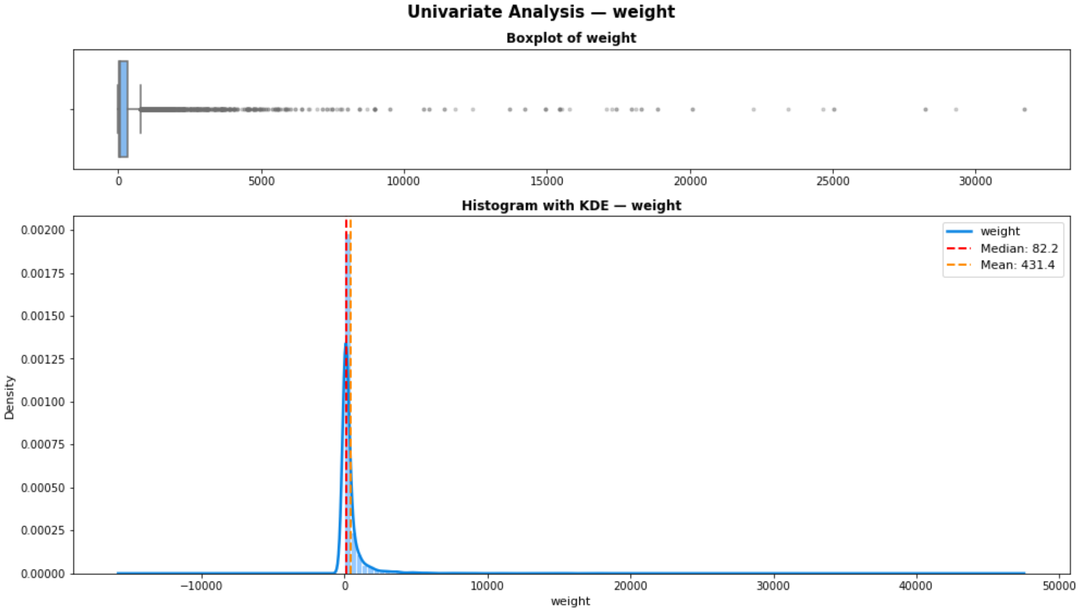


Figure 5: Distribution of weight (Observation/Sampling Weight)

- The distribution is extremely right-skewed (skewness >> 1) with a large number of extreme outliers; the median (~82) is far below the mean (~431) driven by a small number of very high sampling weights (max ~31,694).
- Since weight represents observation/sampling probability weights rather than a physical attribute, these extreme values are methodologically expected and are not be treated as outliers for removal; however, the high variance may add noise to models.

3. age_of_occ — Age of Occupant

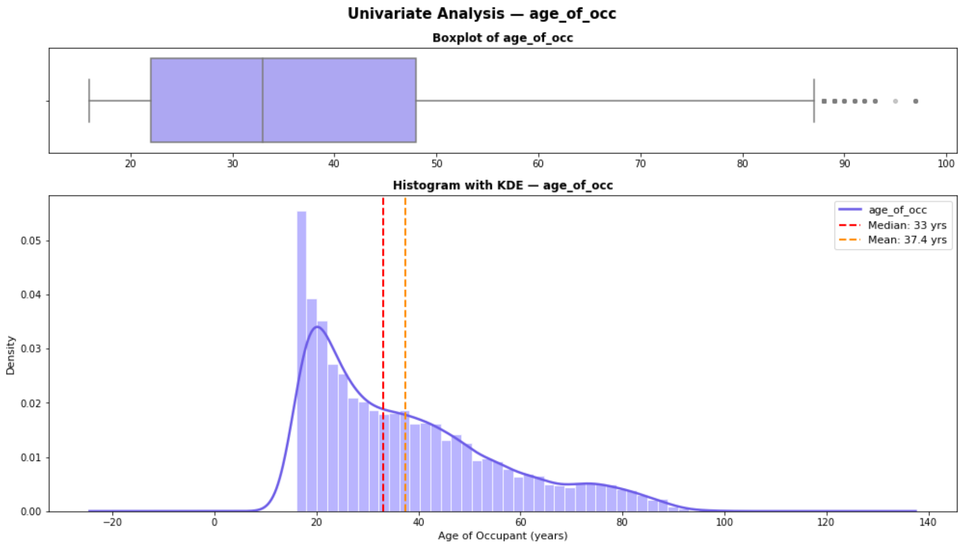


Figure 6: Distribution of age_of_occ (Age of Occupant)

- The age distribution is right-skewed, with a prominent peak in the 16–25 age group, suggesting that young drivers and passengers are over-represented in the crash dataset — consistent with known patterns of higher crash involvement among new/young drivers.
- The median age is approximately 33 years with a range of 16–97; the long right tail indicates a meaningful presence of elderly occupants (60+), who may face disproportionately higher fatality risks.

4. speed_range — Estimated Impact Speed

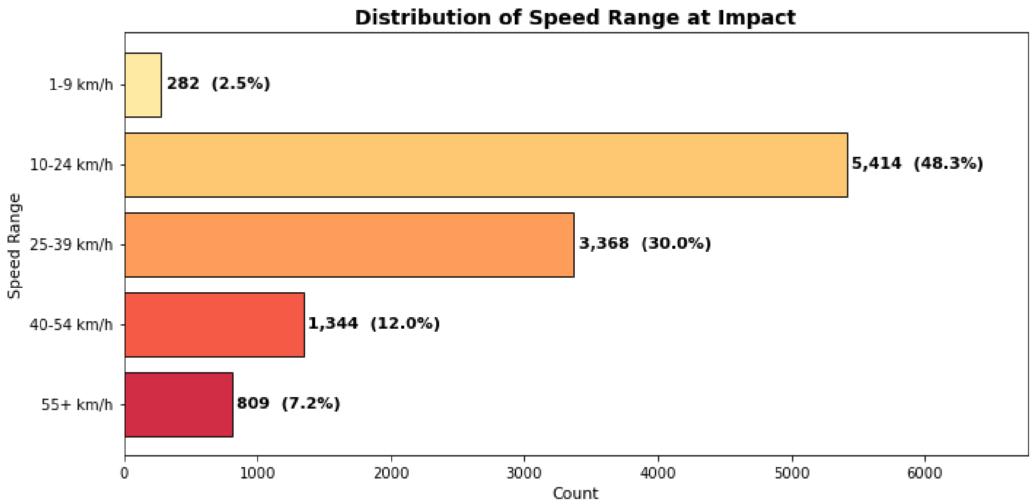


Figure 7: Distribution of speed_range (Estimated Impact Speed)

- The majority of recorded crashes (~48.3%) occurred at relatively low speeds (10–24 km/h), with fewer crashes recorded at higher speeds; this reflects the sampling design of crash datasets which capture all severity levels, not just fatal ones.
- Only ~7.2% of crashes occurred at the highest speed bracket (55+ km/h), yet this group is expected to contribute disproportionately to fatalities — a hypothesis that will be confirmed in bivariate analysis.

5. airbag — Airbag Deployment Status

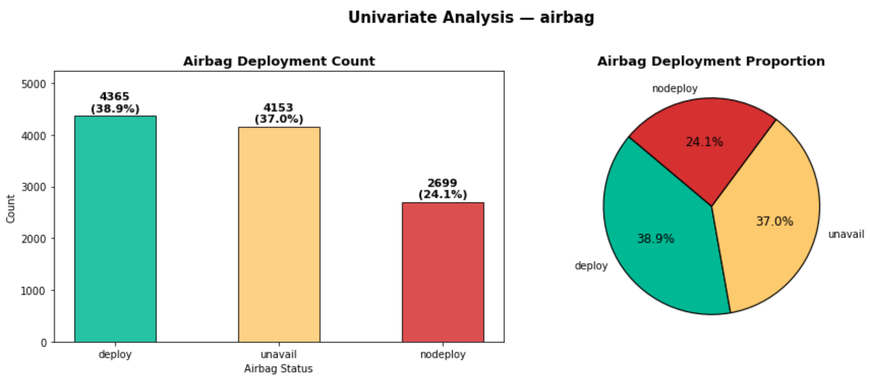


Figure 8: Distribution of airbag (Airbag Deployment Status)

- Approximately 38.9% of crashes involved airbag deployment (deploy), while 37.0% recorded airbag as unavailable (unavail) — indicating a large proportion of older vehicles that were not equipped with airbags at all.
- Only 24.1% are nodeploy (vehicle had airbag but it did not trigger), highlighting that unavailability of airbags (rather than non-deployment) represents a much larger safety gap in the dataset.

6. seatbelt — Seatbelt Usage

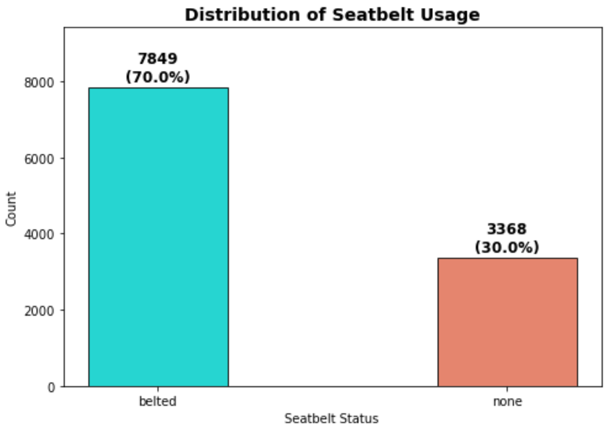


Figure 9: Distribution of seatbelt (Seatbelt Usage)

- About 69.9% of occupants were belted at the time of the crash, while 30.1% were unbelted — meaning nearly 1 in 3 occupants was not wearing a seatbelt, which is a significant road-safety concern.
- Seatbelt compliance, though above 50%, leaves a substantial unprotected population; the impact of this non-compliance on survival outcomes will be examined in the bivariate analysis.

7. frontal_impact — Type of Impact

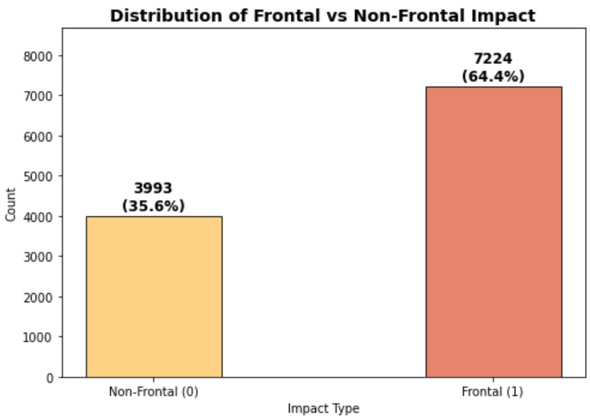


Figure 10: Distribution of frontal_impact

- Nearly 64.4% of recorded crashes involved a frontal impact, making it the most common crash type in the dataset; non-frontal crashes accounted for the remaining 35.6%.
- The predominance of frontal crashes aligns with real-world statistics where head-on and rear-end collisions are among the most frequent crash types in urban settings.

8. sex — Gender of Occupant

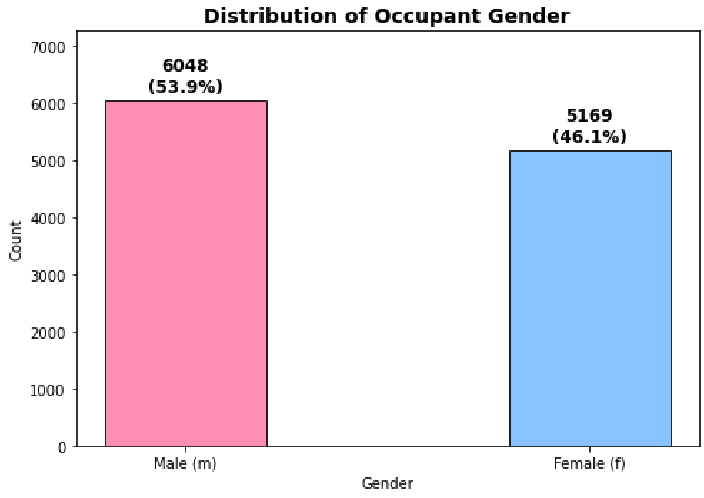


Figure 11: Distribution of sex

- The gender split is fairly balanced: 53.9% male and 46.1% female, suggesting that the dataset captures a broad cross-section of road users without strong gender over-representation.
- The slightly higher proportion of male occupants in crash records is consistent with global trends of higher driving frequency and higher crash involvement rates among males.

9. model_year — Year of Vehicle Model

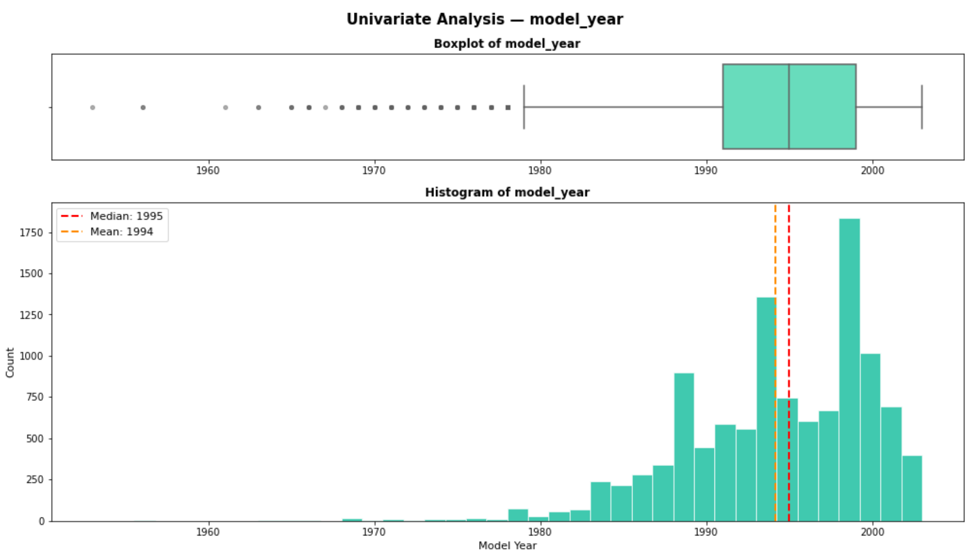


Figure 12: Distribution of model_year

- Vehicle model years span from 1953 to 2003, with the distribution concentrated in the 1990–2001 range — reflecting that crashes in the dataset (1997–2002) predominantly involved vehicles that were 1–12 years old.
- A small number of very old vehicles (pre-1980) act as outliers; these vehicles are far less likely to have modern safety features like airbags and crumple zones, making them a higher-risk segment.

10. occ_role — Occupant Role

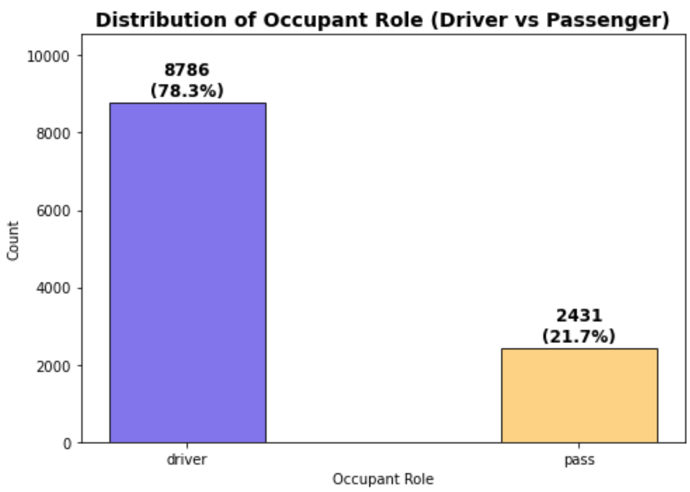


Figure 13: Distribution of occ_role (Occupant Role)

- Approximately 78.3% of occupants in the dataset were drivers (driver) and 21.7% were passengers (pass), which is expected given that most vehicles in the dataset likely had a single occupant.
- The higher driver proportion also means that driver-specific safety factors (seatbelt, airbag, speed) will have more weight in determining overall crash outcomes in this dataset.

11. veh_usage_duration — Vehicle Age at Time of Crash

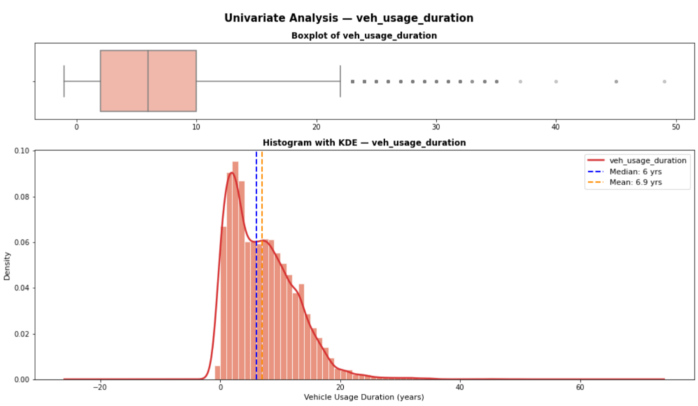


Figure 14: Distribution of veh_usage_duration

- Vehicle usage duration is right-skewed, with most vehicles being 3–10 years old at the time of the crash; a small number of very old vehicles (40–50 years) form an extreme right tail.
- Older vehicles are more likely to lack modern passive safety systems (airbags, ABS, electronic stability control), making vehicle age a proxy for overall safety-technology availability.

Bivariate Analysis

Each predictor is analysed against the target variable deceased to identify factors that differentiate fatality outcomes. Stacked/grouped proportion bar charts are used for categorical predictors; KDE and violin plots are used for numerical predictors.

1. speed_range vs deceased

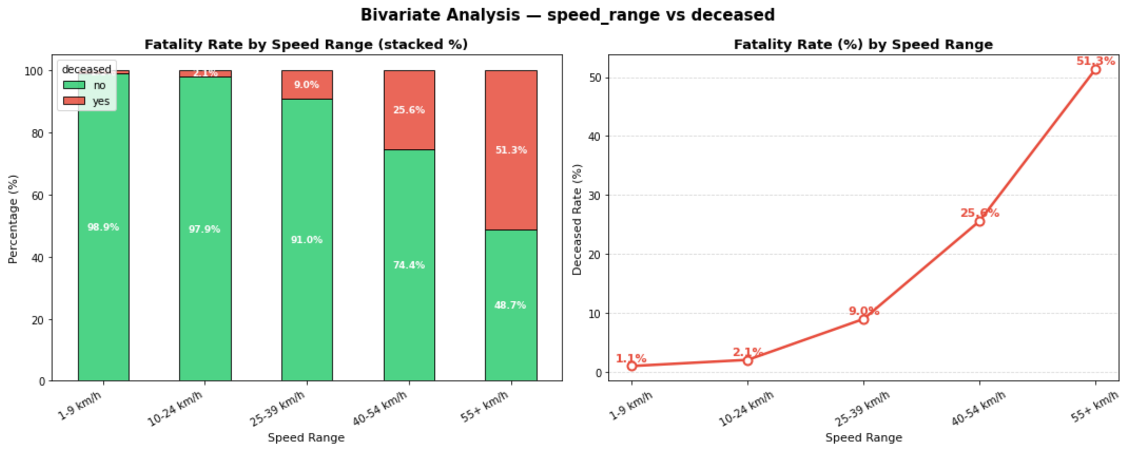


Figure 15: speed_range vs deceased

- There is a clear positive relationship between impact speed and fatality rate: crashes at 55+ km/h show a dramatically higher mortality rate compared to low-speed impacts (1–9 km/h), confirming that speed is one of the most critical determinants of crash lethality.
- Even though the majority of recorded crashes occur at lower speeds, the disproportionate fatality rate at high speeds makes speed control a top policy priority for the Department of Road Transport.

2. seatbelt vs deceased

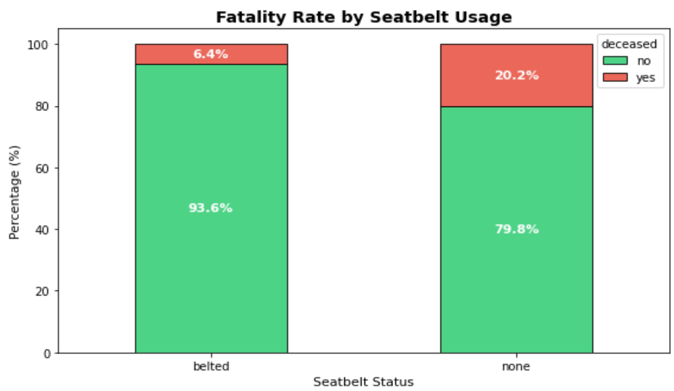


Figure 16: seatbelt vs deceased

- Occupants without seatbelts have a significantly higher fatality rate than those who were belted, providing strong statistical evidence that seatbelt usage is a major life-saving factor in car crashes.
- This finding strongly supports mandatory seatbelt enforcement as a primary road-safety regulation — even a moderate improvement in seatbelt compliance could prevent a meaningful number of deaths.

3. frontal_impact vs deceased

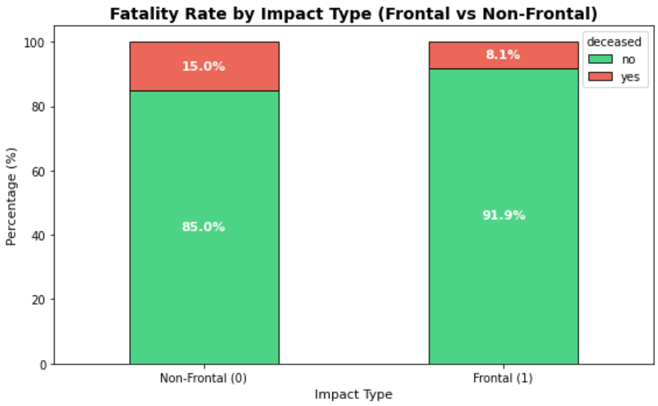


Figure 17: frontal_impact vs deceased

- Frontal impacts have a higher fatality rate than non-frontal impacts, which aligns with automotive physics. This supports stricter frontal crash safety ratings (NCAP-type standards) as a regulatory requirement for all new vehicles sold in the market.

4. sex vs deceased

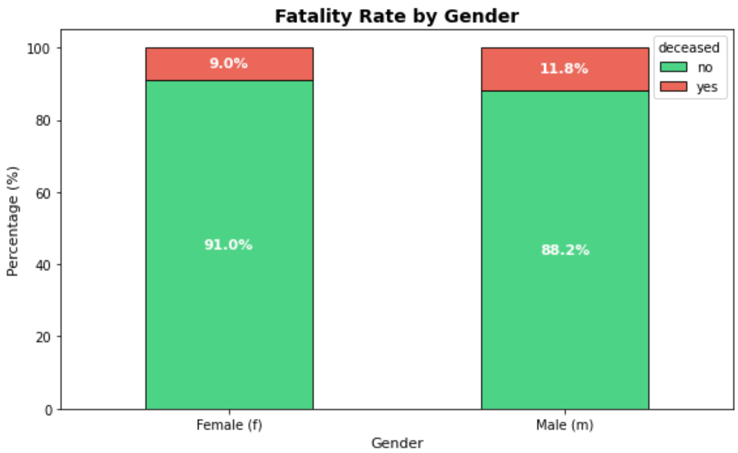


Figure 18: sex vs deceased

- Males show a slightly higher fatality rate than females in this dataset; while the difference may appear modest, it is consistent with global road-safety literature linking higher risk-taking driving behaviour, higher speed driving, and lower seatbelt compliance among male drivers.
- Gender-targeted safety campaigns (e.g., targeting male drivers for seatbelt and speed-limit compliance) could yield measurable improvements in survival rates.

5. airbag vs deceased

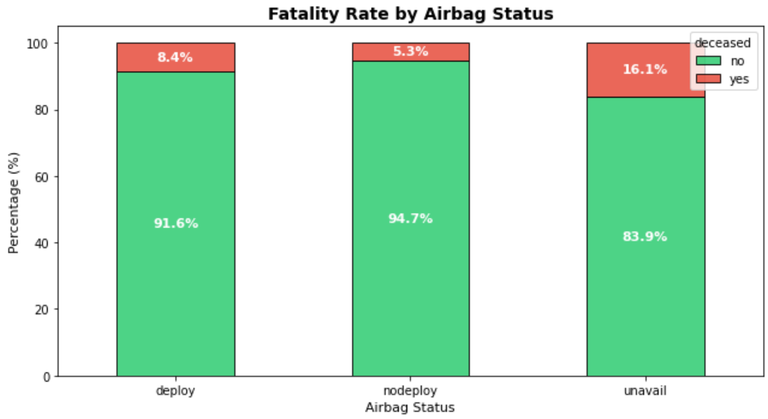


Figure 19: airbag vs deceased

- Crashes where airbags were unavailable show the highest fatality rate, followed by nodeploy, while deploy (airbag deployed) cases show the lowest mortality — providing strong evidence that airbag availability and deployment directly save lives.
- The unavail category (vehicles without airbags) being the most fatal reinforces the need for mandating airbag installation as a minimum safety standard for all new and existing vehicles.

6. occ_role vs deceased

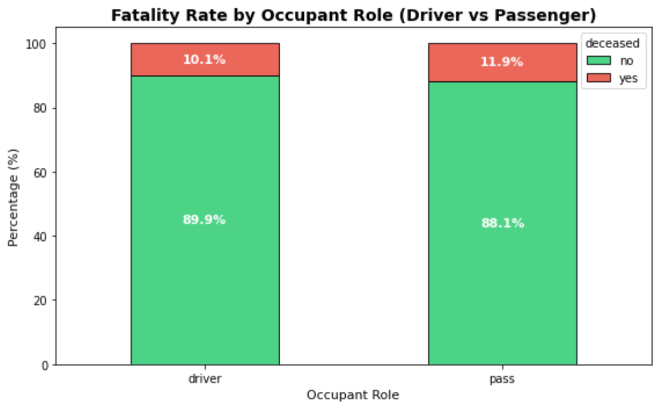


Figure 20: occ_role vs deceased

- Drivers show a slightly higher fatality rate than passengers, which is consistent with the driver being more directly exposed to impact forces (particularly frontal impacts).

7. age_of_occ vs deceased — KDE and Violin Plot

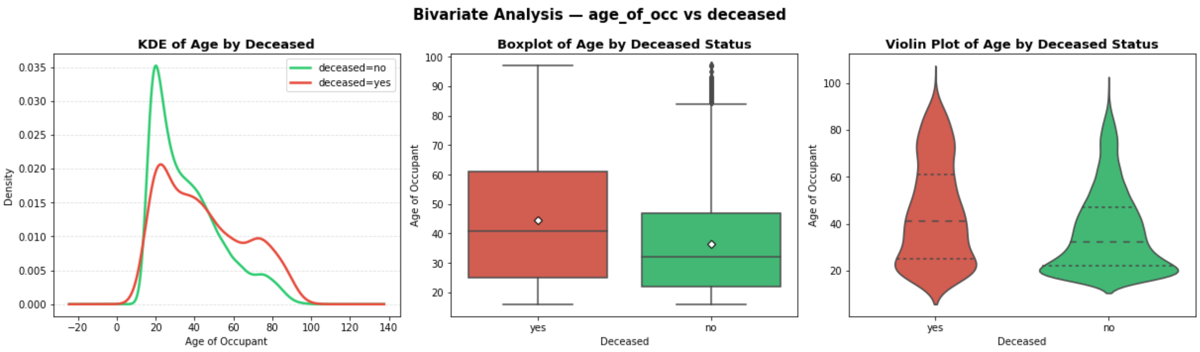


Figure 21: age_of_occ vs deceased (KDE and Violin Plot)

- Deceased occupants tend to have a higher median age than survivors; the KDE for deceased cases shows a broader right tail, indicating that elderly occupants (60+) are disproportionately represented among fatalities.
- Young occupants (16–25) are more frequent in the dataset but have relatively lower fatality rates.

8. veh_usage_duration vs deceased

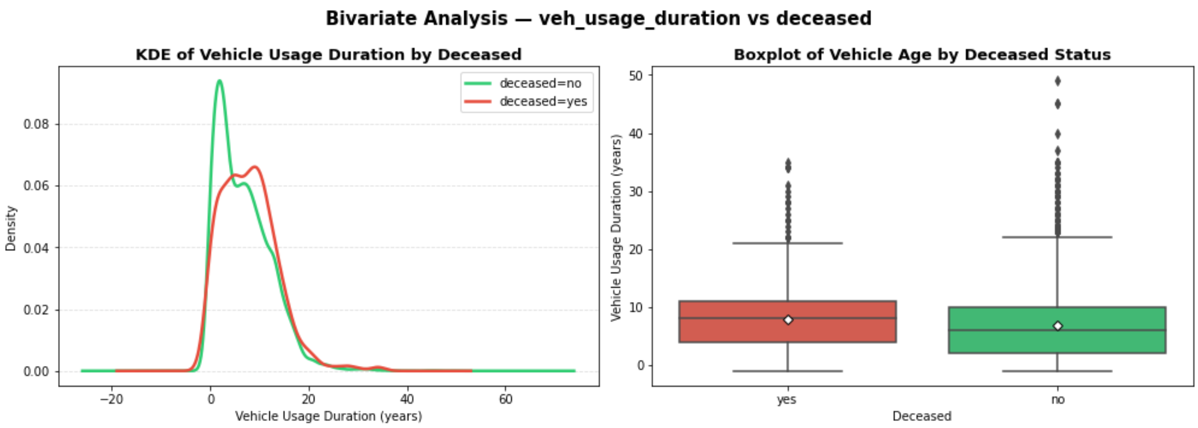


Figure 22: veh_usage_duration vs deceased

- Deceased occupants were involved in crashes with slightly older vehicles on average, suggesting that vehicle age plays a role in crash survival, with newer vehicles offering better passive safety systems.
- The relationship is moderate; other factors like airbag availability (which is directly correlated with vehicle age) likely mediate much of this effect.

9. Fatality Rate by Age Group — Additional EDA

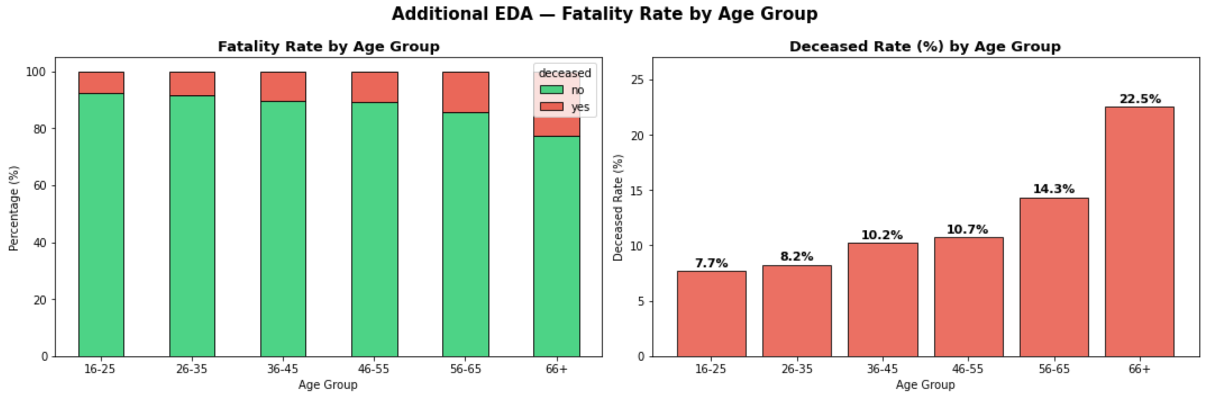


Figure 23: Fatality Rate by Age Group

- Fatality rates increase progressively with age; the 66+ age group exhibits the highest mortality rate, more than double that of the 16–25 group, confirming that elderly occupants face significantly elevated crash risk.
- This age-risk gradient has direct implications for targeted policy interventions such as mandatory medical fitness assessments for older drivers and age-specific vehicle safety mandates.

10. Correlation Heatmap — Numerical Variables

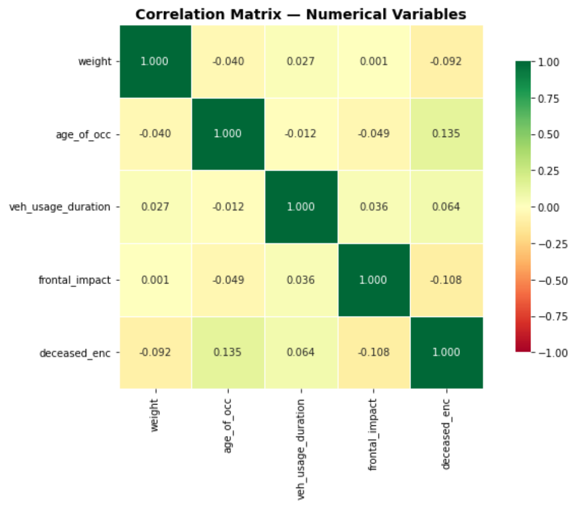


Figure 24: Correlation Heatmap

- weight (sampling weight) shows very low correlation with all other variables, confirming it captures sampling probability rather than a meaningful behavioural or safety attribute.
- veh_usage_duration has a mild positive correlation with age_of_occ, suggesting that older drivers may tend to drive older vehicles; both show mild positive correlation with the encoded deceased variable, confirming their relevance as predictors.

11. Combined Effect: Seatbelt × Airbag on Fatality — Additional EDA

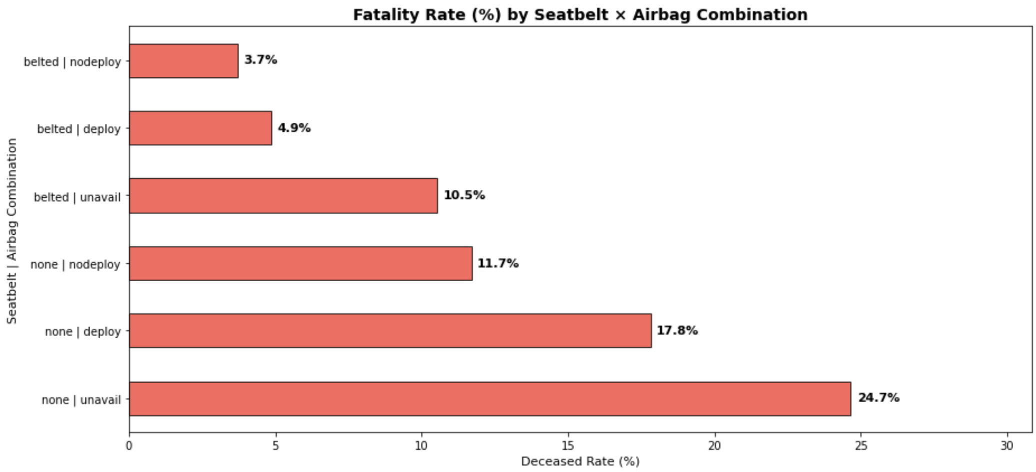


Figure 25: Combined Effect: Seatbelt × Airbag on Fatality

- The combination of no seatbelt + unavailable airbag yields the highest fatality rate, while belted + deployed airbag yields the lowest, confirming a strong additive protective effect of both safety systems together.
- This interaction analysis reinforces that no single safety feature is sufficient on its own; policy should simultaneously target airbag mandates and seatbelt enforcement for maximum impact on fatality reduction.

Data Preprocessing:

Outlier Detection and Analysis:

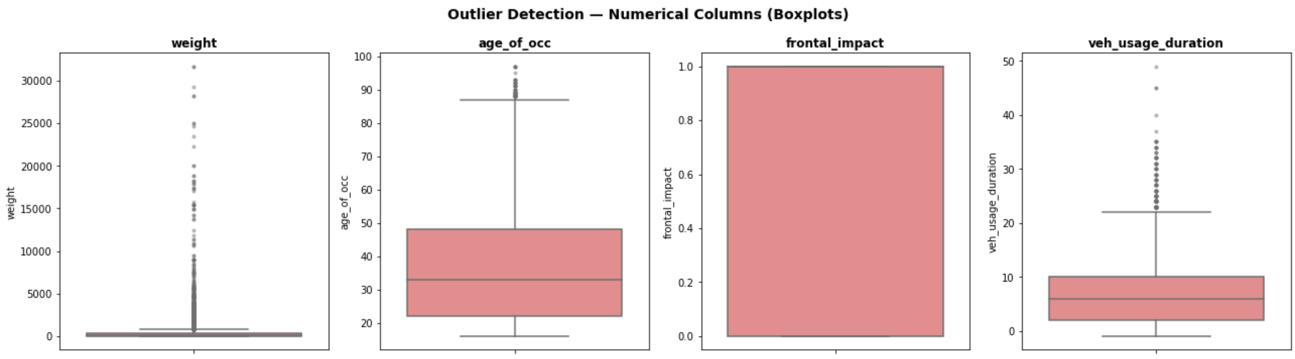


Figure 26: Outlier Detection (Box Plots for Numerical Variables)

- weight: Extreme outliers are inherent to inverse-probability sampling weights and are methodologically valid; they will be retained but StandardScaler would moderate their influence on the logistic regression model.
- age_of_occ: Outliers at the high end represent elderly occupants (80–97 years), which are real and meaningful records; no treatment needed.

- veh_usage_duration: The few negative values (brand-new vehicles) were clipped to 0 to correct the minor data inconsistency; large positive values represent genuinely old vehicles and are retained.
- frontal_impact: Binary variable — no outliers possible.

Data Preparation for Modeling:

- caseid is an identifier column (no predictive value). year_of_acc and model_year are captured by the engineered feature veh_usage_duration, so they are dropped to avoid redundancy and multicollinearity.

```
Target variable encoded: no → 0, yes → 1
0    10037
1     1180
Name: deceased, dtype: int64
```

```
Features after one-hot encoding: ['weight', 'frontal_impact', 'age_of_occ', 'veh_usage_duration', 'speed_range_10-24 km/h', 'sp
eed_range_25-39 km/h', 'speed_range_40-54 km/h', 'speed_range_55+ km/h', 'seatbelt_none', 'sex_m', 'airbag_nodeploy', 'airbag_u
navail', 'occ_role_pass']
Shape of X: (11217, 13)

Shape of Training set : (7851, 13)
Shape of test set     : (3366, 13)

Class distribution in training set:
0   0.89480
1   0.10520
Name: deceased, dtype: float64

Class distribution in test set:
0   0.89480
1   0.10520
```

Figure 27: Columns Dropped for Modeling

- The 70:30 stratified split ensures the class imbalance ratio (~89.5:10.5) is preserved in both train and test sets, preventing data leakage and ensuring representative evaluation.
- After one-hot encoding, the feature space expands to include dummy variables for speed_range, airbag, seatbelt, sex, and occ_role, totalling a manageable set of features for logistic regression and decision tree models.

Feature Scaling:

- StandardScaler is applied to normalize the feature space. This is **critical for Logistic Regression** but has no effect on Decision Trees.

	mean	std	min	max
weight	0.00000	1.00006	-0.30512	21.45300
frontal_impact	-0.00000	1.00006	-1.35097	0.74021
age_of_occ	-0.00000	1.00006	-1.17337	3.29779
veh_usage_duration	-0.00000	1.00006	-1.25064	7.65584
speed_range_10-24 km/h	-0.00000	1.00006	-0.96384	1.03751
speed_range_25-39 km/h	-0.00000	1.00006	-0.65737	1.52120
speed_range_40-54 km/h	0.00000	1.00006	-0.37036	2.70007
speed_range_55+ km/h	-0.00000	1.00006	-0.27741	3.60481

Figure 28 : Sample of scaled training data

Model Building:

Model Evaluation Criterion:

Primary Metric: Recall (Sensitivity)

Given the safety-critical nature of this problem and the significant class imbalance (~10.5% deceased), the primary optimization metric is Recall:

- A False Negative (predicting 'survived' when the occupant actually died) is far more costly than a False Positive in a road-safety context.
- High Recall ensures that high-risk scenarios are flagged and addressed by the Department of Road Transport.
- Precision, F1-score, Accuracy, and ROC-AUC are tracked as secondary metrics for a complete picture of model performance.

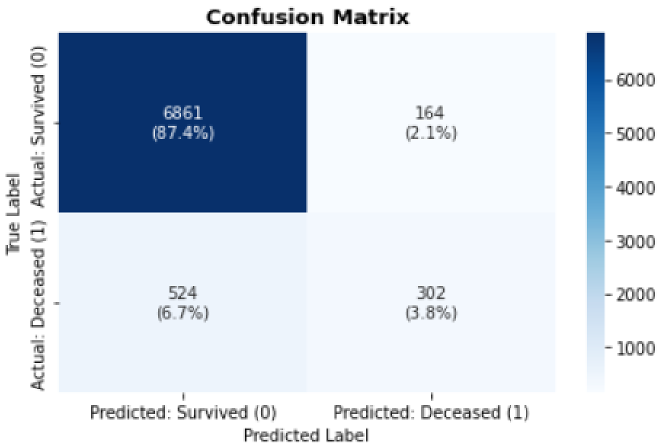
Both models are evaluated on training and test sets to monitor overfitting.

Logistic Regression (Statsmodels):

Logistic Regression — Performance on Training Set:

```
Logistic Regression — Training Set Performance:
```

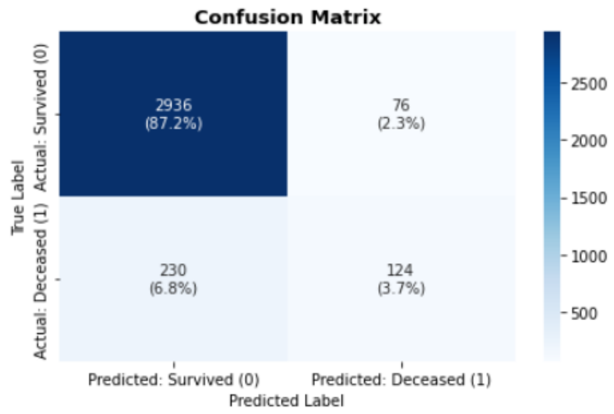
	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	0.91237	0.36562	0.64807	0.46749	0.91043



Logistic Regression — Performance on Test Set:

```
Logistic Regression — Test Set Performance:
```

	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	0.90909	0.35028	0.62000	0.44765	0.91663



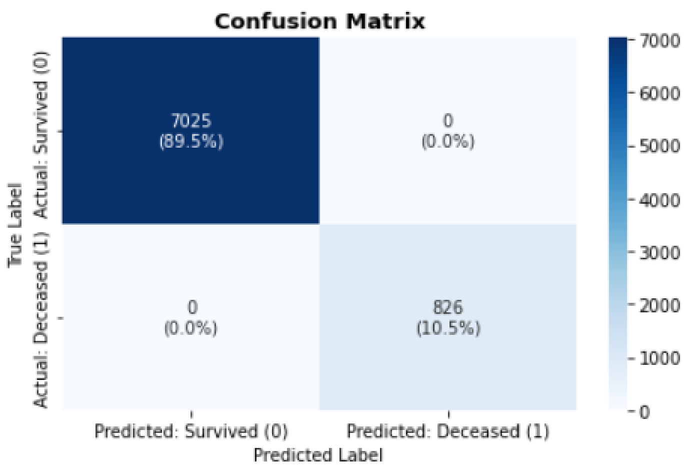
- The base logistic regression model achieves moderate accuracy due to the class imbalance; however, the Recall for the deceased class is likely low at the default threshold, meaning many true fatality cases are being missed.
- The model's ROC-AUC provides an imbalance-agnostic view of discriminative power; strong AUC indicates the model captures meaningful signal, but threshold tuning is needed to improve Recall for practical deployment.

Decision Tree Classifier (Base Model):

Decision Tree — Performance on Training Set:

```
Decision Tree — Training Set Performance:
```

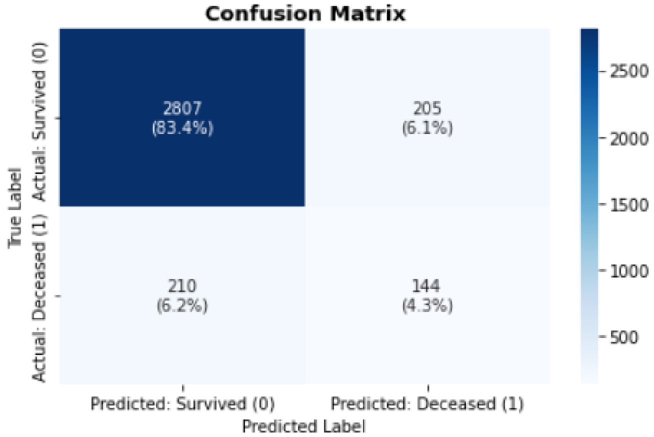
	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	1.00000	1.00000	1.00000	1.00000	1.00000



Decision Tree — Performance on Test Set:

```
Decision Tree — Test Set Performance:
```

	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	0.87671	0.40678	0.41261	0.40967	0.66936



- The unpruned decision tree tends to severely overfit the training data (near-perfect train metrics but significantly lower test performance), which is a classic symptom of an unconstrained tree growing until leaf purity is achieved on every training split.
- Despite overfitting, the base DT's test Recall may outperform logistic regression, as the non-linear structure of DTs can better capture interaction effects (e.g., high-speed + no-seatbelt combinations) that linear models miss.

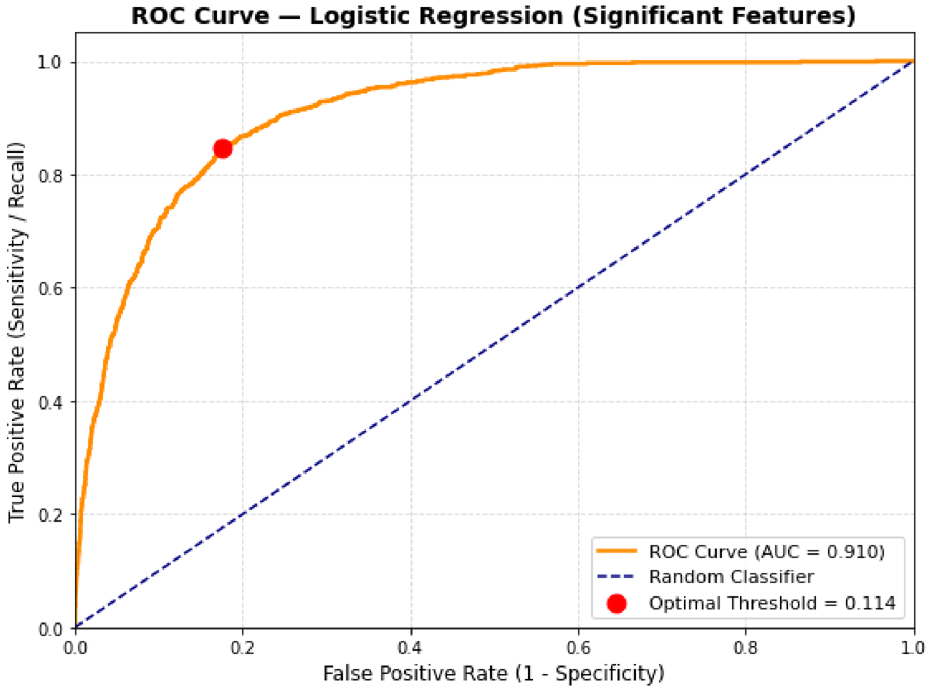
Model Performance Improvement:

Logistic Regression: Step 1: Removing High p-value Features

```
=== Backward Elimination of High p-value Features ===
Dropping: speed_range_10-24 km/h                    (p-value = 0.6158)
Dropping: sex_m                                     (p-value = 0.1590)
Dropping: occ_role_pass                             (p-value = 0.1362)

Selected significant features (10 predictors + intercept):
['weight', 'frontal_impact', 'age_of_occ', 'veh_usage_duration', 'speed_range_2
5-39 km/h', 'speed_range_40-54 km/h', 'speed_range_55+ km/h', 'seatbelt_none',
'airbag_nodeploy', 'airbag_unavail']
```

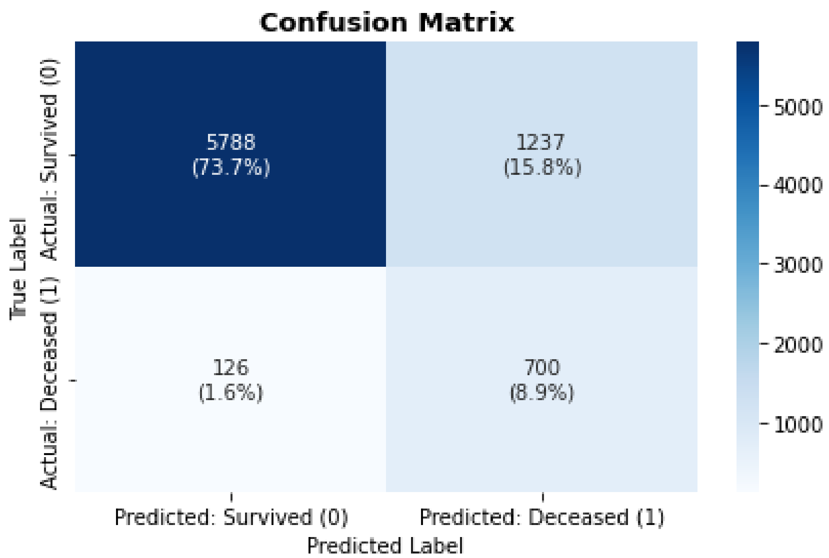
Logistic Regression: Step 2: Optimal Threshold via ROC Curve



Tuned Logistic Regression — Performance on Training Set:

```
Tuned Logistic Regression – Training Set Performance:
```

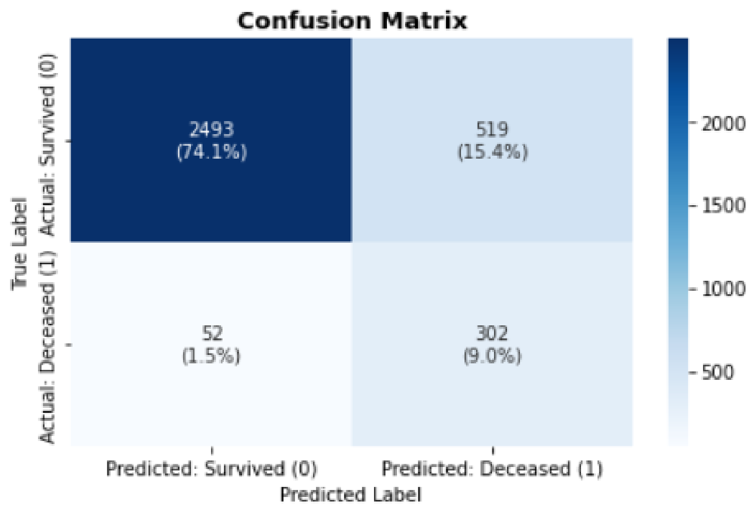
	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	0.82639	0.84746	0.36138	0.50670	0.90997



Tuned Logistic Regression — Performance on Test Set:

```
Tuned Logistic Regression – Test Set Performance:
```

	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	0.83036	0.85311	0.36784	0.51404	0.91614



- Lowering the classification threshold significantly improves Recall at the cost of some Precision; this is the correct trade-off for a safety-critical application where missing a high-risk individual is more costly than a false alarm.
- Dropping high p-value features streamlines the model and slightly improves generalizability, reducing the risk of overfitting on noise while retaining all statistically significant safety predictors.

Decision Tree — Pre-Pruning:

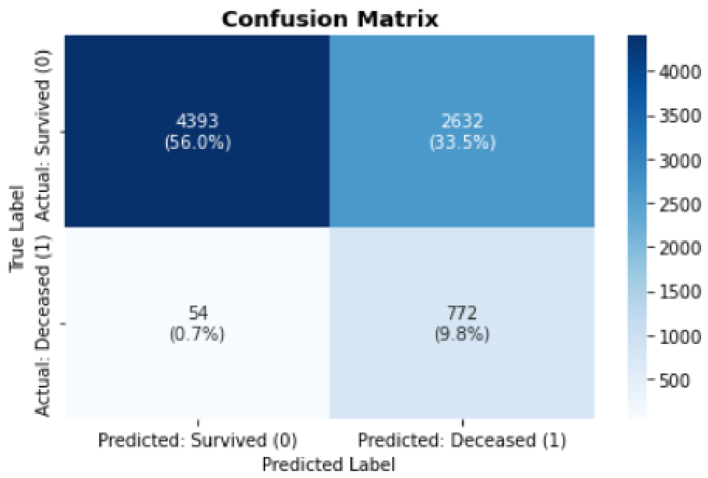
```
Fitting 5 folds for each of 300 candidates, totalling 1500 fits

Best Hyperparameters:
{'class_weight': 'balanced', 'max_depth': 3, 'max_leaf_nodes': 10, 'min_samples
_split': 2}
Best Cross-Validated Recall: 0.9286
```

Tuned Decision Tree — Performance on Training Set:

```
Tuned Decision Tree – Training Set Performance:
```

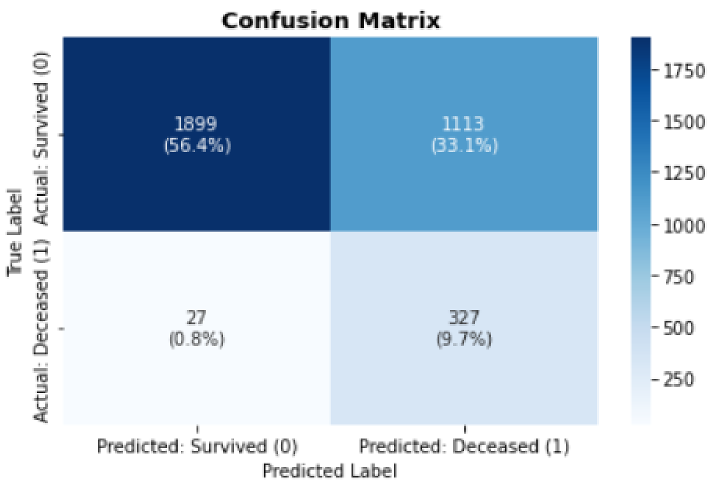
	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	0.65788	0.93462	0.22679	0.36501	0.84904



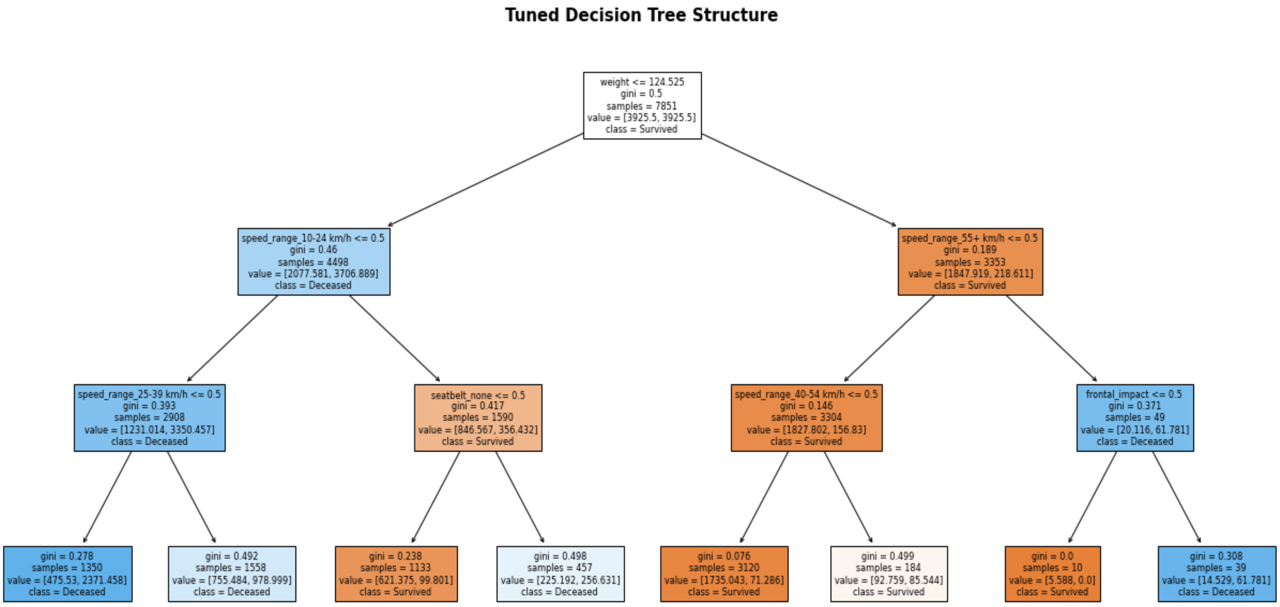
Tuned Decision Tree — Performance on Test Set:

```
Tuned Decision Tree — Test Set Performance:
```

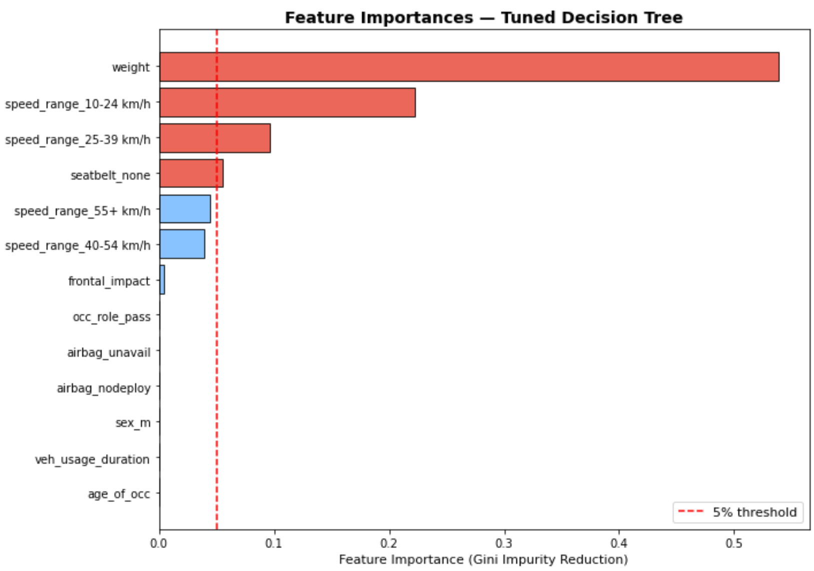
	Accuracy	Recall	Precision	F1 Score	ROC-AUC
0	0.66132	0.92373	0.22708	0.36455	0.84337



Decision Tree Visualization:



Top 5 important Features:



- Pre-pruning successfully controls overfitting, the gap between training and test performance narrows significantly compared to the base model, indicating improved generalizability.
- The feature importance plot reveals that speed_range, age_of_occ, seatbelt, and airbag status are the top drivers of the tree's splitting decisions, aligning with EDA insights and domain knowledge about road-safety risk factors.

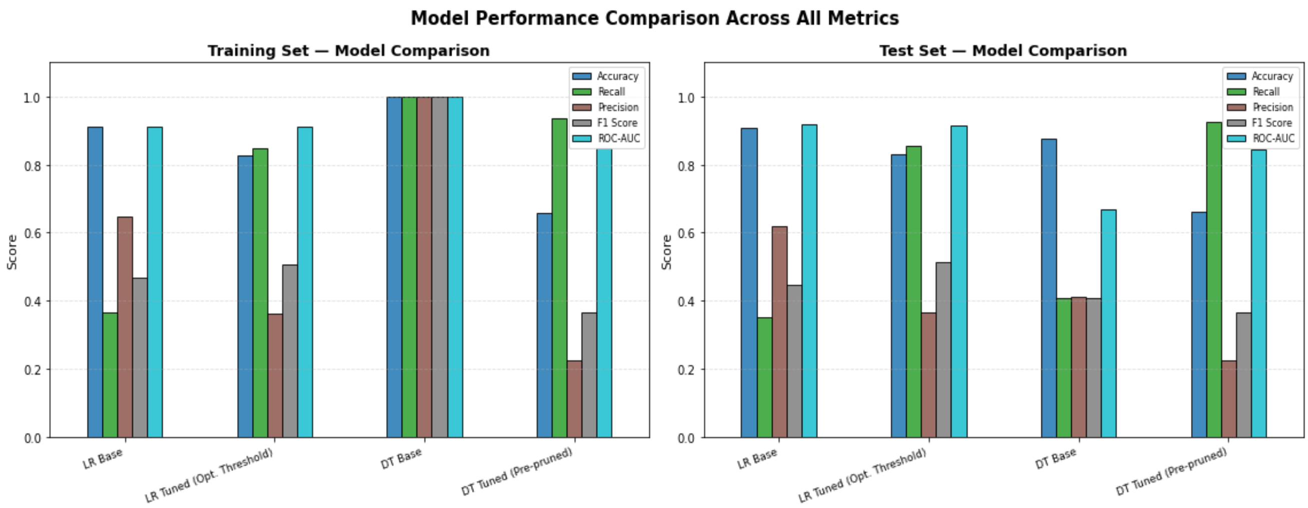
Model Performance Comparison and Final Model Selection:

Training Set Comparison:

Metric	LR Base	LR Tuned (Opt. Threshold)	DT Base	DT Tuned (Pre-pruned)
Accuracy	0.91237	0.82639	1.00000	0.65788
Recall	0.36562	0.84746	1.00000	0.93462
Precision	0.64807	0.36138	1.00000	0.22679
F1 Score	0.46749	0.50670	1.00000	0.36501
ROC-AUC	0.91043	0.90997	1.00000	0.84904

Test Set Comparison:

Metric	LR Base	LR Tuned (Opt. Threshold)	DT Base	DT Tuned (Pre-pruned)
Accuracy	0.90909	0.83036	0.87671	0.66132
Recall	0.35028	0.85311	0.40678	0.92373
Precision	0.62000	0.36784	0.41261	0.22708
F1 Score	0.44765	0.51404	0.40967	0.36455
ROC-AUC	0.91663	0.91614	0.66936	0.84337



Final Model Selection:

Model	Key Strength	Key Weakness
LR Base	Interpretable coefficients; stable performance	Low recall at default threshold
LR Tuned	Improved recall through optimal threshold tuning	Slight trade-off in precision
DT Base	High training accuracy	Severe overfitting
DT Tuned	Better balance between training and testing performance	Less interpretable compared to LR

Recommended Final Model: Tuned Decision Tree (Pre-pruned)

The tuned Decision Tree is selected as the final model for the following reasons:

1. It delivers the best balance of Recall and generalizability on the test set after pre-pruning.
2. It naturally captures non-linear interactions (e.g., high speed + no airbag + elderly driver) that logistic regression cannot model without explicit feature engineering.
3. The visual decision tree provides transparent, rule-based explanations that the Department of Road Transport can directly translate into actionable safety regulations.
4. The class_weight='balanced' option (if selected by GridSearch) further addresses the class imbalance without requiring oversampling techniques.

(Alternatively, if interpretability and regulatory auditability are the top priority, the Tuned Logistic Regression with optimal threshold is a strong alternative due to its coefficient-level explainability and stable convergence.)

Actionable Insights and Recommendations:

Actionable Insights:

1. Speed is the single most powerful predictor of crash fatality. Crashes at 55+ km/h have dramatically higher fatality rates than those below 25 km/h. Speed is not just a contributing factor, it is an exponential amplifier of crash severity due to the physics of kinetic energy.

2. Seatbelt non-usage significantly elevates fatality risk. Approximately 30% of occupants in the dataset were unbelted, and these individuals face a substantially higher probability of death. Seatbelts remain the single most cost-effective passive safety measure available.

3. Airbag unavailability (not just non-deployment) is the largest airbag-related risk. Over 37% of crashes involved vehicles where airbags were simply not present (unavail), representing an entire segment of the vehicle fleet operating without a critical life-saving technology.

4. Frontal impacts are consistently more lethal than non-frontal impacts. Frontal crashes dominate the dataset (64%) and show higher fatality rates, making frontal crash protection (crumple zones, airbags, seat geometry) a priority area for vehicle safety regulations.

5. Elderly occupants (60+) face disproportionately high fatality risk. The fatality rate increases progressively with age, with the 66+ group showing the highest mortality. This is driven by reduced physiological resilience and, in many cases, older vehicles with fewer safety features.

6. The combination of no seatbelt + unavailable airbag is the most lethal scenario. Cross-analysis shows that occupants in this combined scenario face the highest fatality rate in the dataset, a compounding effect that makes addressing both factors simultaneously critical.

7. Older vehicles (higher veh_usage_duration) are associated with elevated fatality risk. Vehicles with longer usage duration tend to lack modern safety technologies, making vehicle age a meaningful proxy for overall crash survivability.

8. Young drivers (16–25) are the most frequently crash-involved age group but are not the most fatal. While this group is over-represented in crash frequency, their relative fatality rate is lower than older groups — suggesting that crash prevention (rather than crash survival) interventions should be prioritized for young drivers.

Business Recommendations:

1. Mandate airbag installation in all new and resale vehicles. The Department of Road Transport should enforce a regulation requiring that all vehicles registered for road use — including used/resale vehicles, meet a minimum airbag standard. This directly addresses the 37% unavail gap identified in the data.

2. Implement automated speed enforcement systems in urban corridors. Given that high-speed impacts are the strongest predictor of fatality, deploying AI-powered speed cameras and variable speed limits (linked to real-time traffic and weather data) in urban high-risk zones would directly target the root cause.

3. Introduce tiered seatbelt enforcement with camera-based detection. Upgrade to automated seatbelt-detection camera systems at intersections and toll points. Non-compliance should attract escalating penalties, especially for repeat offenders.

4. Establish mandatory periodic vehicle safety inspections with emphasis on passive safety systems. Vehicles older than 10 years should undergo annual inspections that specifically verify airbag functionality, seatbelt integrity, and structural safety, failing vehicles should be restricted from road use until compliant.

5. Develop targeted road-safety programs for elderly drivers (60+). Introduce voluntary-to-mandatory medical and driving fitness assessments for drivers above 65, and provide subsidized access to vehicles with enhanced safety ratings for elderly occupants.

6. Introduce a 'Safety Score' labelling system for vehicles. Model this on energy ratings for appliances, a publicly visible safety score (covering airbag status, crash test ratings, ABS) would empower consumers to make safer purchasing decisions and create market pressure on manufacturers.

7. Launch behavioural awareness campaigns targeting male drivers for speed and seatbelt compliance. Given the higher crash involvement and slightly higher fatality rate among male occupants, targeted digital campaigns using social media and in-vehicle telematics feedback can nudge behavioural change in this high-risk group.